

Sales Force Automation (SFA) System

QA Handover Documentation

Prepared for: Quality Assurance & Operations Team

Document Type: Project Functional, Setup, API and Test Guide

Version: 1.0

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Project Area	Details
Application Type	Role-based sales force automation, field force tracking, multi-branch inventory, sales orders and delivery handover management web application.
Frontend	React + Vite + TailwindCSS + Recharts
Backend	Node.js + Express.js (Modular Monolith)
Database	Neon PostgreSQL with Prisma ORM
Primary Users	Super Admin, Company Admin, Head of Sales, Sales Manager, Sales Executive, Inventory Manager, Warehouse Manager

Note: This document is prepared for QA handover. Actual test credentials, staging URLs, and secret environment values must be filled by the development team before sharing with external stakeholders.

1. Document Control

Field	Value
Project Name	Sales Force Automation (SFA) System
Document Name	QA Handover Documentation
Version	1.0
Prepared For	QA Team & Stakeholders
Prepared By	Development Team
Date	19 August 2026
Status	Draft for QA execution

This document explains the project purpose, roles, business flow, functional modules, API overview, database overview, setup steps, QA test cases, known issues, and bug reporting format.

2. Project Overview

The Sales Force Automation (SFA) System is a role-based web application designed to manage the complete operational sales cycle—from sales order creation and field task assignment to multi-warehouse inventory stock checks, customer signature capture, warehouse handover verification, and sales order fulfillment.

The system separates responsibilities by role so that each user sees only the modules required for their work. The main business flow starts from Admin-created operational users and continues through Head of Sales, Sales Manager, Sales Executive, Inventory Manager, and Warehouse Manager.

Business Area	Purpose
Sales Order Management	Create customer sales orders, track order status (Draft, Submitted, Approved, Completed), line item calculation, and stock availability checks.
Inventory & Warehouse	Create warehouses, manage product catalogs, track stock levels (Physical, Reserved, Net Available), and approve or reject order handover requests.
Field Force & Delivery	Assign delivery tasks, track field sales executive progress, capture customer digital signatures upon handover, and verify location logs.
Target & Operational Performance	Track role-based operational metrics: Sales Executive Task Execution Rate % and Sales Manager Order Fulfillment Rate % alongside real-time sales order revenue trends.
Company & Branch Admin Control	Create branches, departments, teams, and manage organization user accounts with branch-level scoping.
Super Admin Control	Create Company Admins and view high-level multi-organization analytics and platform health.

3. Scope for QA

QA should validate functional correctness, role-based access control (RBAC), data consistency, UI behavior, error handling, and complete end-to-end workflow.

In Scope	Out of Scope / Conditional
Login, logout, refresh token and protected routes.	Load testing and advanced performance testing unless specifically requested.
Role-wise dashboards and sidebar navigation.	Production payment gateway integrations if not configured.
Manager, Executive, and Warehouse user account creation and RBAC limits.	Real hardware GPS device testing unless mobile/HTTPS environment is enabled.
Sales Order creation to warehouse pickup to delivery signature handover flow.	Third-party SMS gateway notifications if mock email/logger is active.
Warehouse stock movement, net available quantity calculations, and low stock alerts.	Advanced barcode/QR scanning hardware integrations unless explicitly enabled.

4. Technology Stack

Layer	Technology	QA Notes
Frontend	React + Vite + TailwindCSS + Recharts	Run using <code>npm run dev</code> from frontend folder. Validate routes, forms, charts, and API integration.
Backend	Node.js + Express.js	Run using <code>npm run dev</code> from backend folder. Validate REST APIs, role guards, and error handling.
Database	Neon PostgreSQL + Prisma ORM	Use Prisma Client v7+. Direct SQL / Prisma Queries must respect organization and branch scoping.
Authentication	JWT + Refresh Token	Validate session persistence, profile picture persistence, expired token behavior, and protected routes.
Analytics & Charts	Recharts library	Validate that 0-export analytics pages render live DB figures for Area, Line, and Bar charts.

5. User Roles and Responsibilities

Role	Main Responsibility	Access Level
Super Admin	Creates Company Admins and views platform-wide system analytics.	Platform-level
Company Admin	Manages organization structure: branches, departments, teams, and user accounts.	Organization-level
Head of Sales	Monitors sales revenue trajectory, Target & Performance dashboard, and 5-tab operational reports.	Strategic Sales Level
Sales Manager	Manages branch operations, assigns tasks to sales executives, and tracks branch sales orders.	Branch-level
Sales Executive	Executes field tasks, logs customer visits, captures digital delivery signatures, and creates orders.	Field Operations
Inventory Manager	Manages product catalogs, multi-warehouse stock allocations, and stock movements.	Global Logistics
Warehouse Manager	Operates specific warehouse console, handles stock check-in/out, and approves pickup handovers.	Warehouse Level

6. End-to-End Business Flow

The primary flow QA should test is shown below:

Company Admin creates Sales Manager -> Sales Manager creates Sales Order & Assigns Task -> Field Sales Executive picks up stock from Warehouse -> Executive completes delivery & captures Customer Signature -> Warehouse Manager approves Handover Verification -> Order completes, invoice generates & stock deducts.

Step	Role	Action	Expected Result
1	Super Admin	Create Company Admin	Company Admin user is created and can log in.
2	Company Admin	Create Branches, Managers & Executives	Users are created with correct role and branch mapping.
3	Sales Manager / Exec	Create Sales Order	Order appears in Sales Orders list in Draft/Submitted state.
4	Sales Manager	Assign Task to Sales Executive	Task appears in Sales Executive's assigned task console.
5	Sales Executive	Pick up stock & Deliver to Customer	Executive captures customer digital signature on mobile UI.
6	Warehouse Manager	Approve Delivery Handover	Handover is approved; stock deducts automatically & invoice issues.
7	Head of Sales	View Target & Performance Page	Charts update with real DB task execution % and order fulfillment %.

7. Role-wise Functional Documentation

Super Admin

- Login and redirect to Super Admin Dashboard.
- Create Company Admin accounts.
- View platform-level multi-tenant analytics and organization counts.
- Access User Management to view all organization users (supports limit=1000 without page truncation).
- Super Admin does not manage daily operational branch tasks directly.

Company Admin

- Login and redirect to Company Admin Dashboard.
- Create organization structure: Branches, Departments, Teams, and User accounts.
- Assign users to specific branches and roles.
- Manage Company Admin profile and credentials.

Head of Sales

- Login and redirect to Head of Sales Strategic Leadership Dashboard.
- View Organization Sales Order Revenue & Trajectory Area Chart with Daily, Weekly, and Monthly timeframe toggles.
- View Target & Operational Performance Analytics (Sales Executive Task Execution % & Sales Manager Order Fulfillment %).
- View 5-Tab Reports: Branch Performance, Warehouses, Products, Stock per Branch, and Field Force Workforce.
- Sidebar navigation contains only strategic modules (Dashboard, Sales Orders, Target & Performance, Reports, Notifications, Settings).

Sales Manager

- View Branch Manager Dashboard with branch-scoped sales orders, tasks, and team stats.
- Create and manage Sales Orders for branch customers.
- Assign delivery and field tasks to branch Sales Executives.
- View team member detail pages and performance scores.

Sales Executive

- View Sales Executive Dashboard with assigned tasks and scheduled visits.
- Execute assigned field tasks and log customer visit outcomes.
- Capture customer digital delivery signatures directly on mobile/web interface.
- Submit delivery handover proof to Warehouse Manager.

Inventory Manager

- View multi-warehouse stock dashboard, product catalogs, and price lists.
- Create new warehouses with capacity details.
- Create products and assign inventory allocations to warehouses.
- Track global stock movements, stock adjustments, and low stock warnings.

Warehouse Manager

- View Warehouse Manager Console for assigned warehouse.
- Process stock check-in and check-out entries.
- Review and approve/reject delivery handover requests submitted by field executives.

- Approving handover automatically deducts reserved stock and triggers completed invoice status.

8. Sidebar & Module Navigation Matrix

QA should verify that each role sidebar contains only authorized working sections without duplicate or restricted links.

Role	Sidebar Sections	QA Validation
Super Admin	Dashboard, Organization, Sales Orders, Target & Performance, Reports, Settings	Verify all 24 organization users load in Users list.
Company Admin	Dashboard, Organization (Branch, Dept, Team, Users), Settings	Verify branch and team creation forms work cleanly.
Head of Sales	Dashboard, Sales Orders, Target & Performance, Reports, Settings	Verify restricted Organization/Team links are hidden.
Sales Manager	Dashboard, Sales Orders, Team Management, Settings	Verify data is scoped to user's branchId.
Sales Executive	Dashboard, Field Force (My Tasks, My Visits), Settings	Verify signature pad modal works on task submission.
Inventory Manager	Dashboard, Inventory (Products, Warehouses, Stock, Movements)	Verify net available quantity math is accurate.
Warehouse Manager	Dashboard, My Warehouse Console, Stock, Handover Approvals	Verify handover approval deducts physical stock.

9. Order Execution & Warehouse Handover Flow

Flow Stage	Expected Data & UI Behavior
Sales Order Creation	Customer name, order items, quantities, unit prices, total amount, and status set to Draft/Submitted.
Stock Reservation	Order items reserve stock in designated warehouse. Net Available Qty = Physical Qty - Reserved Qty.
Field Delivery Task	Task assigned to Sales Executive with pickup warehouse address and delivery destination.
Handover & Signature	Field Executive uploads digital signature proof. Task moves to Handover Verification Pending.
Warehouse Approval	Warehouse Manager approves handover. Reserved stock is cleared, physical stock deducts, and invoice issues.

10. Setup and Installation Guide

QA should run the backend and frontend separately. Execute commands from the repository root folders:

Backend Setup:

```
cd backend
npm install
npm run dev
```

Frontend Setup:

```
cd frontend
npm install
```



```
npm run dev
```

Service	Default Local URL	QA Check
Backend API	http://localhost:5000	Server should start without crash and connect to Neon PostgreSQL.
Frontend App	http://localhost:5173	Login page should load and connect to backend APIs.
Database	Neon PostgreSQL	Backend should connect using DATABASE_URL from .env file.

11. Environment Variables

```
PORT=5000
NODE_ENV=development
DATABASE_URL=your_neon_database_url
DIRECT_URL=optional_direct_neon_url_for_schema_scripts
JWT_SECRET=your_jwt_secret
REFRESH_TOKEN_SECRET=your_refresh_token_secret
CLIENT_URL=http://localhost:5173
```

Variable	Purpose	QA Note
DATABASE_URL	Main PostgreSQL connection string	Must point to valid Neon DB.
DIRECT_URL	Direct DB connection for migrations	Useful for schema ALTER scripts.
JWT_SECRET	Signs access tokens	Required for login authentication.
REFRESH_TOKEN_SECRET	Signs refresh tokens	Required for session refresh flow.
CLIENT_URL	Frontend URL for CORS	Must match running frontend port.

12. API Endpoint Overview

Module	Method	Endpoint	Purpose
Auth	POST	/api/v1/auth/login	User authentication & JWT token.
Auth	GET	/api/v1/auth/me	Get current user profile.
Users	GET	/api/v1/users?limit=100	Fetch organization users list.
Users	PUT	/api/v1/users/profile	Update user profile & avatar.
Sales Orders	POST	/api/v1/sales-orders	Create new sales order.
Sales Orders	GET	/api/v1/sales-orders	List sales orders.
Targets	GET	/api/v1/target-performance/company-overview	Get target analytics & trends.
Reports	GET	/api/v1/reports/organization-analytics	Get 5-tab operational reports.
Inventory	GET	/api/v1/inventory/warehouses	List warehouse master records.
Inventory	GET	/api/v1/inventory/stock	Get stock levels per warehouse.
Tasks	POST	/api/v1/team/tasks	Create delivery task.
Tasks	PATCH	/api/v1/team/tasks/:id/handover	Submit delivery signature proof.

13. Database Entity Overview

Entity/Table	Purpose	Important Fields / Notes
users	Stores user accounts and credentials.	id, email, avatarUrl, branchId, isActive.
organizations	Multi-tenant root organization entity.	id, name, code.
branches	Branch office locations.	id, name, code, latitude, longitude.
warehouses	Warehouse locations and capacity.	id, name, capacity, address, branchId.
products	Master product catalog.	id, name, sku, price, unit.
stocks	Inventory stock records per product/warehouse.	quantity, reservedQuantity, availableQuantity.
orders	Customer sales order records.	orderNumber, totalAmount, status, ownerId.
tasks	Field delivery tasks.	assignedToId, status, signatureUrl, dueDate.

14. Test Credentials Template

Role	Email	Password	Expected Redirect
Super Admin	devendradangi9174@gmail.com	To be provided	Super Admin Dashboard
Company Admin	lakshya123@gmail.com	To be provided	Company Admin Dashboard
Head of Sales	kashish123@gmail.com	To be provided	Head of Sales Dashboard
Sales Manager	mayur.rudrawal@itsoftlab.com	To be provided	Manager Dashboard
Sales Executive	darshan123@gmail.com	To be provided	Sales Dashboard
Inventory Manager	anurag123@gmail.com	To be provided	Inventory Dashboard
Warehouse Manager	honey123@gmail.com	To be provided	Warehouse Console

15. QA Test Cases

Module	Test Case	Steps	Expected Result
Authentication	Valid Login	Login with valid role credentials.	User redirects to correct role dashboard.
User Management	Fetch All Users	Open Organization -> Users page as Super Admin.	All 24 users load without truncation.
Sales Orders	Create Sales Order	Fill order form with items and submit.	Order appears in orders list with total amount.
Target & Perf.	View Target Graphs	Open Target & Performance page.	Recharts plot real DB task % and order fulfillment %.
Reports	5-Tab Navigation	Click through Branch, Warehouse, Product, Stock, Field Force tabs.	All tabs load real DB metrics with 0 export buttons.
Field Delivery	Digital Signature	Submit task handover with digital signature.	Signature saves and handover submits to warehouse.
Warehouse	Approve Handover	Warehouse Manager approves handover.	Physical stock deducts and invoice completes.

16. Role-Based Access Control Test Matrix

Feature	Super Admin	Company Admin	Head of Sales	Sales Manager	Sales Exec	Warehouse Mgr
Create Company Admin	Allowed	Denied	Denied	Denied	Denied	Denied
Create Branch / User	Denied	Allowed	Denied	Denied	Denied	Denied
View Target Analytics	Allowed	Read-only	Allowed	Branch-only	Denied	Denied
Create Sales Order	Denied	Read-only	Denied	Allowed	Allowed	Denied
Digital Signature Handover	Denied	Denied	Denied	Denied	Allowed	Denied
Approve Stock Handover	Denied	Denied	Denied	Denied	Denied	Allowed

17. Live Tracking & Field Force Testing Notes

Live tracking and visit logs depend on browser location permission and GPS availability.

Check	Expected Behavior
Location Permission Denied	App shows clean permission prompt and does not crash.
Task Execution Signature	Canvas saves base64 PNG signature to backend storage.
Network Disconnect	App queues offline task state or alerts user.

18. Bug Reporting Format

Bug Title:

Module:

Role:

Environment:

Browser/Device:

Steps to Reproduce:

Expected Result:

Actual Result:

Console Error:

API Error Response:

Severity: Critical / High / Medium / Low

Reported By:

Severity	Meaning	Example
Critical	Blocks core workflow or app crashes.	User cannot login; order flow blocked.
High	Major feature broken but workaround exists.	Handover approval fails for valid task.
Medium	Feature works partially or UI issue.	Customer name missing in table.
Low	Minor UI/typo/alignment issue.	Button spacing or label typo.

19. Final QA Handover Checklist

- Frontend runs successfully using `npm run dev`.
- Backend runs successfully using `npm run dev`.
- Database connection connects cleanly to Neon PostgreSQL.
- All test users (Super Admin, Company Admin, Head of Sales, Sales Managers, Sales Executives, Inventory Managers, Warehouse Managers) are provided.
- Role-wise dashboard redirection works for all 7 roles.
- Super Admin can view all 24 organization users without page truncation.
- Head of Sales sidebar contains only strategic links and renders real DB trend charts.
- Sales Executive can execute task and capture digital delivery signature.
- Warehouse Manager can approve handover verification, deducting physical stock automatically.

End of Document